

SQL ON TWO TABLES

id last_name first_name DoB Income Genre

1 Lopez Baranda Christina 15/11/2000 55000 Fantasy 2 Jin-Soon Sin 29/03/1983 65000 Crime 3
Jones Hannah 01/02/1973 129000 Fantasy 4 Novak Stanislaw 12/12/1992 91000 Crime 5 Turay
Tandice 09/07/1980 99000 Romance 6 Roy Shanta 11/10/1977 55000 Fantasy 7 Berger Henry
15/08/1956 63000 Romance 8 Khatami Paree 11/10/1966 86000 Sci-Fi

id title ISBN

1 Creating relational databases for fun and profit 7654321123456 2 Relational databases for really smart people 9876543212345 3 My life with relational databases: a memoir 3212345678909 4 Relational databases: an existential journey 8172635412345

book_id author_id

3	6
2	4
2	5
1	1
1	3
1	5
4	8

edition_id book_id date_of_publication edition_number

1	3	2001	1
2	3	2003	2
3	4	2003	1
5	1	2000	1
6	3	2005	3
8	2	2012	1
9	3	2009	4

edition_id book_id date_of_publication edition_number

1	3	2001	1
2	3	2003	2
3	4	2003	1
5	1	2000	1
6	3	2005	3
8	2	2012	1
9	3	2009	4

- Each edition is related to a book
- book_id is a **foreign key** that refers to books
- Each *non null* values of book_id must be found in the id column of books

What if a book is deleted or its id changed?

We have 3 choices, decided when the constraint is created:

1. delete/change also the edition (CASCADE)
2. abort the operation
3. set the book_id to NULL (SET NULL)

All choices guarantee the integrity of the database

Books

id title ISBN

1 Creating relational databases for fun and profit 7654321123456 2 Relational databases for really smart people 9876543212345

BooksAuthors

book_id author_id

2	4
2	5
1	1

This query asks for the first and last names of authors of the book with id 1:

The results are:

```
first_name last_name
```

Hannah Jones

Christina Lopez Baranda Tandice Turay

```
SELECT first_name , last_name  
FROM Authors , BooksAuthors  
WHERE BooksAuthors.author_id = Authors.id  
AND book_id = 1;
```

```
SELECT first_name , last_name  
FROM Authors JOIN BooksAuthors  
      ON BooksAuthors.author_id = Authors.id  
WHERE book_id = 1;
```

The JOIN version is better when only 2 tables are involved

To find the book IDs and ISBNs that have editions published after (that is, greater than) 2003.

id ISBN date_of_publication

2 9876543212345 2012 3 3212345678909 2005 3 3212345678909 2009

```
SELECT Books.id, ISBN, date_of_publication
FROM Books, Editions
WHERE Books.id = Editions.book_id
AND Editions.date_of_publication > 2003;
```

id	title	ISBN
----	-------	------

2 Relational databases for really smart people 9876543212345 3 My life with relational databases: a memoir 3212345678909 3 My life with relational databases: a memoir 3212345678909

```
SELECT id , title , ISBN
FROM Books, Editions
WHERE Books.id = Editions.book_id
AND Editions.date_of_publication > 2003;
```

1. Duplicate rows

id	title	ISBN
----	-------	------

2 Relational databases for really smart people 9876543212345 3 My life with relational databases: a memoir 3212345678909

```
SELECT DISTINCT id, title, ISBN
FROM Books, Editions
WHERE Books.id = Editions.book_id
AND Editions.date_of_publication > 2003;
```

Find who has written a book whose ISBN ends with 5

id ISBN id last_name first_name

2 9876543212345 4 Novak Stanislaw 2 9876543212345 5 Turay Tandice 4 8172635412345 8 Khatami
Paree

```
SELECT Books.id , ISBN , Authors.id ,  
       last_name , first_name  
FROM Books , Authors , BooksAuthors  
WHERE Books.id = BooksAuthors.book_id AND  
       BooksAuthors.author_id = Authors.id AND  
       isbn LIKE "%5";
```

```
SELECT Books.id , ISBN , Authors.id ,  
       last_name , first_name  
FROM Books , Authors , BooksAuthors  
WHERE Books.id = BooksAuthors.book_id AND  
       BooksAuthors.author_id = Authors.id AND  
       isbn LIKE "%5";
```

1. There are three tables involved
2. id is a column name of two tables, use the table to disambiguate

Find who has a last name with exactly 5 characters and has written a book whose ISBN ends with 5

id ISBN id last_name first_name

2 9876543212345 4 Novak Stanislaw 2 9876543212345 5 Turay Tandice

```
SELECT Books.id, ISBN, Authors.id, last_name, first_name
FROM Books, Authors, BooksAuthors
WHERE Books.id = BooksAuthors.book_id AND
       BooksAuthors.author_id = Authors.id AND
       last_name LIKE '_____' AND isbn LIKE "%5";
```

```
SELECT Books.id, ISBN, Authors.id, last_name, first_name
FROM Books, Authors, BooksAuthors
WHERE Books.id = BooksAuthors.book_id AND
       BooksAuthors.author_id = Authors.id AND
       last_name LIKE '_____' AND isbn LIKE "%5";
```

```
SELECT b.id, ISBN, a.id, last_name, first_name
FROM Books b, Authors a, BooksAuthors ba
WHERE b.id = ba.book_id AND
       ba.author_id = a.id AND
       last_name LIKE '_____' AND isbn LIKE "%5";
```

```
SELECT T1.A  
FROM T1, T2  
WHERE T1.B=T2.C;
```

1. Cross product between T1 and T2
2. Select only the rows satisfying the WHERE clause
3. Projection on the A column

For each author, list the ISBN of the books they have written

```
SELECT Books.id , ISBN , Authors.id , last_name , first_name
FROM Books , Authors , BooksAuthors
WHERE Books.id = BooksAuthors.book_id AND
       BooksAuthors.author_id = Authors.id;
```

But the author with id 2 has written no books

id ISBN id last_name first_name

3 3212345678909 6 Roy Shanta 2 9876543212345 4 Novak Stanislaw 2 9876543212345 5 Turay Tandice 1
7654321123456 1 Lopez Baranda Christina 1 7654321123456 3 Jones Hannah 1 7654321123456 5 Turay
Tandice 4 8172635412345 8 Khatami Paree

```
SELECT Authors.id, last_name, first_name, Books.id, ISBN  
FROM Books, Authors LEFT JOIN BooksAuthors  
    ON Authors.id=BooksAuthors.author_id  
WHERE Books.id = BooksAuthors.book_id;
```

id last_name first_name id ISBN

6 Roy Shanta 3 3212345678909 4 Novak Stanislaw 2 9876543212345 5 Turay Tandice 2 9876543212345 1
Lopez Baranda Christina 1 7654321123456 3 Jones Hannah 1 7654321123456 5 Turay Tandice 1
7654321123456 8 Khatami Paree 4 8172635412345 2 Jin-Soon Sin NULL NULL 7 Berger Henry NULL
NULL

For each author, find the number of books they have written

```
SELECT Authors.id , last_name , first_name ,  
       count(*) as number  
FROM Books , Authors , BooksAuthors  
WHERE Books.id = BooksAuthors.book_id AND  
       BooksAuthors.author_id = Authors.id  
GROUP BY Authors.id ;
```

For each author, find the number of books they have written

```
SELECT Authors.id , last_name , first_name ,  
       count(*) as number  
FROM Books , Authors , BooksAuthors  
WHERE Books.id = BooksAuthors.book_id AND  
       BooksAuthors.author_id = Authors.id  
GROUP BY Authors.id ;
```

NO

The variables in the SELECT and in the GROUP BY clauses must be consistent

For each author, find the number of books they have written

```
SELECT Authors.id , last_name , first_name ,  
       count(*) as number  
FROM Books , Authors , BooksAuthors  
WHERE Books.id = BooksAuthors.book_id AND  
       BooksAuthors.author_id = Authors.id  
GROUP BY Authors.id , last_name , first_name ;
```

```
id last_name first_name number
```

6 Roy Shanta 1 4 Novak Stanislaw 1 5 Turay Tandice 2 1 Lopez Baranda Christina 1 3 Jones Hannah 1
8 Khatami Paree 1

For each author, find the number of books they have written

```
SELECT Authors.id, last_name, first_name ,  
       count(Books.id) as number  
FROM Books, Authors LEFT JOIN BooksAuthors  
     ON Authors.id=BooksAuthors.author_id  
WHERE Books.id = BooksAuthors.book_id  
GROUP BY Authors.id, last_name, first_name ;
```

count(Books.id) counts the number of tuples where Books.id is not NULL

```
id last_name first_name number
```

6 Roy Shanta 1 4 Novak Stanislaw 1 5 Turay Tandice 2 1 Lopez Baranda Christina 1 3 Jones Hannah 1
8 Khatami Paree 1 2 Jin-Soon Sin 0 7 Berger Henry 0

Find the authors that have written no book

```
SELECT Authors.id, last_name, first_name ,  
FROM Authors LEFT JOIN BooksAuthors  
    ON Authors.id=BooksAuthors.author_id  
WHERE BooksAuthors.book_id IS NULL;
```

Comparisons with NULL values require IS NULL or IS NOT NULL

```
id last_name first_name number
```

2 Jin-Soon Sin 7 Berger Henry

Except where noted, text and images for Introduction to Relational Databases by Mark Jordan is licensed under a Creative Commons Attribution 4.0 International License. Except where noted, text and images for Introduction to Relational Databases by Gianluca Della Vedova is licensed under a Creative Commons Attribution 4.0 International License. Everything in the 'scripts' directory is in the public domain (CC0).